

NoBroker: Using AI to match homeowners and home seekers with the end-to-end services they need

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ABOUT NOBROKER



NOBROKER.COM

Rent | Buy | Sell

About NoBroker

NoBroker helps its customers make estate decisions Google Cloud provides services such as Storage, Google Maps Engine, Google Maps

NoBroker helps customers make smarter real estate decisions, with the help of artificial intelligence. With algorithms that connect property owners directly with buyers and tenants, it saves customers \$150 million in broker fees per year. NoBroker now offers a suite of real estate services from rent payment to visitor management on its digital portal, powered by Google Cloud.

Industries: Real Estate & Construction

Platform, and Cloud AI.

Location: India

Google Cloud results

- Achieves 99.9% uptime by autoscaling on Google Kubernetes Engine
- Facilitates crowdsourcing of 35,000 listings a month with Cloud Vision AI
- Connects 77% more owners and renters by dynamically updating pages with latest available properties to keep content fresh

Reduced main supply by 80%

Cloud Vision API

Google Cloud (<https://cloud.google.com/>) Google

Maps Platform

Cloud Storage (<https://cloud.google.com/storage/>)

Directions API

Compute Engine

(<https://cloud.google.com/compute/>)

Distance Matrix

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Geocoding API

Cloud Vision API (<https://cloud.google.com/vision/>)

tation/geocoding/intro

Google Maps Platform

(<https://cloud.google.com/maps-platform/>)

tation/android-sdk/intro

Directions API

(<https://developers.google.com/maps/documentation/directions/start>)

Maps JavaScript API

Distance Matrix

(<https://developers.google.com/maps/documentation/distance-matrix/start>)

Geocoding API

(<https://developers.google.com/maps/documentation/geocoding/intro>)

Maps Android SDK

(<https://developers.google.com/maps/documentation/android-sdk/intro>)

Maps iOS SDK

(<https://developers.google.com/maps/documentation/ios-sdk/intro>)

Maps JavaScript API

(<https://developers.google.com/maps/documentation/javascript/tutorial>)

Google Places API

(<https://cloud.google.com/maps-platform/places/>)

- Enables development of Touchless Entry using facial recognition in just three weeks, for contact-free security, protecting residents' hygiene during COVID-19

The real estate industry in India is undergoing a digital transformation. Real estate companies are investing in new technologies such as AI, cloud services, and mapping platforms to address changing trends and customer demands. NoBroker (<https://www.nobroker.com>) reinvents the real estate experience with an AI-driven platform to link owners to buyers and tenants directly, without the middleman.

High broker fees and duplicate listings are common pain points for property owners and home seekers in India. Some brokers charge as much as 10 months' worth of rent or 6% of a property sale in broker fees. Customers save on brokerage fees by using NoBroker's app for end-to-end property services from listing to closing.

“We’re running AI algorithms on Google Cloud (<https://cloud.google.com/>) to match customers with top properties that meet their requirements,” said Akhil Gupta, co-founder and CTO of NoBroker.com. “Our AI applications will continuously learn from new datasets and deliver more accurate results for property owners, home buyers, and tenants.”

Having established its presence over six years, NoBroker is now focused on improving its services using innovative technologies. It delivers the benefits of machine learning to its customers through AI projects such as Smart Recommendations for quality of life indicators, Rent-o-meter for rent predictions, Refer-And-Earn for crowdsourcing property listings, and Touchless Entry for facial recognition security features.

The Smart Recommendations project helps customers to find their ideal place to live. Using Google Maps Platform ([/maps-platform](https://maps-platform)) as a data source, the algorithm searches for local amenities and public transit and assigns a Livability score and Transit score. Such information complements property details from owners and helps buyers and tenants to make informed decisions about each property.

Meanwhile, Rent-o-meter (<https://www.nobroker.in/know-your-rent/>) runs on Google Cloud to help property owners and tenants estimate rent prices, an invaluable tool for NoBroker.com, where rentals make up 85% of transactions. The custom algorithm predicts rent with 95% accuracy by analyzing historical

transaction data, property images, market value, trends, and the neighborhood. Landlords are more likely to attract tenants by setting the right rental price. NoBroker helps property owners save time by auto-generating a well-written property description to highlight key selling points.

NoBroker has also developed and launched a feature called Touchless Entry that uses facial recognition to detect residents of a property, including household staff. Two times faster than fingerprint biometrics, Touchless Entry takes just 200 milliseconds to detect a face. Most importantly, it aims to deliver safe, secure, and fast entry. Brought to market in just three weeks, as India went into lockdown due to COVID-19, Touchless Entry aimed to protect residents by helping them reduce physical contact with surfaces where the virus might live. NoBroker's timely release was made possible by leveraging Google Cloud AI/ML products combined with TensorFlow (<https://www.tensorflow.org/>).

Finally, the Cloud Vision (/vision)-powered () program uses crowdsourcing to generate 35,000 property listings every month. Anyone using the NoBroker app can upload a photo of a "to let" or "for sale" board to the platform. The optical character recognition (OCR) feature on Cloud Vision AI extracts the owner's phone number from the photo and correlates it with listing user information on NoBroker's MySQL database. It's a win-win for everyone. The finder earns approximately \$1.35 for each verified new listing while the owner gets more visibility for their property.

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and CTO, NoBroker.com

Keeping up with growth and compliance on Google Cloud

In 2019, NoBroker announced plans to expand its broker-free platform across the top twenty cities in India within three to four years. With the company's growing presence, more customers access the app online and on mobile devices. The platform captures close to a billion clickstream messages daily.

Before moving to Google Cloud, NoBroker experienced challenges with its network, which was taking up a lot of resources and bandwidth. It also required several people to manage, which meant those employees weren't able to add value to the business in other areas. From a business perspective, revenue suffered whenever the network was down.

NoBroker administrators often scrambled to troubleshoot the performance bottlenecks. At times it was Network File System (NFS) connection failure due to the legacy host server in Singapore. Such situations took the service provider hours to resolve. On other occasions, NoBroker needed more capacity to cope with incoming traffic.

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“We meet compliance requirements by storing data in the Google Cloud region in India. From a security standpoint, the biggest benefit of Google Cloud is that our data sits in a virtual private cloud. We can effectively limit risk exposure to public IP addresses and endpoints.”

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At the same time, NoBroker needed to address government regulation around customer data. India

introduced a Personal Data Protection Bill in 2019 that requires companies to store sensitive personal data in local servers to protect users.

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Google Kubernetes Engine helps to reduce admin overheads

In the past, scaling resources on the legacy system was a manual process for the engineering and DevOps team that required 45 minutes to an hour. An administrator had to spin up a VM, write an image, install applications, and run the VM at night when the impact on traffic is low.

“There’s always a risk of configuration errors with manual provisioning,” says Akhil. “Auto-provisioning on [Google Kubernetes Engine](#) (/kubernetes-engine) removes manual errors by adding and removing Kubernetes clusters in a timely and consistent manner.”

“We reduced 80% of maintenance support time needed by moving to Google Cloud,” Akhil adds.

“Automating tasks with Google Kubernetes Engine allows developers to devote time to improving user experience and features instead of maintaining the infrastructure.”

NoBroker set CPU and memory limits on Google Cloud to make sure a container never runs out of memory. “If CPU usage goes above 80%, we spin up a new machine. If utilization falls below 50%, we scale down the resource to save server costs,” Akhil elaborates. “During the COVID-19 situation, for example, we were able to scale down with ease.”

“Google Kubernetes Engine is ideal for running applications such as Elasticsearch at massive scale,” says Akhil. “We deploy around 800 cores of Elasticsearch at 2.4 terabytes of RAM on Google Cloud so customers can get property search results in milliseconds.”

On top of that, NoBroker uses Google Kubernetes Engine for continuous integration and deployment (CI/CD) to speed up development time. CI/CD pipelines run automated workflows such as code checks and make changes if necessary. It’s easy for developers to build, test, and deploy applications at scale on Google Kubernetes Engine without worrying about configuration. Developers create a Kubernetes cluster and deploy production-ready code at the click of a button.

Accelerating AI development for innovation with Cloud GPUs

In addition to existing AI projects, NoBroker’s data science team is now running a machine learning project on Google Cloud to improve customer interactions. NoBroker is transcribing one million hours of call center conversations (100,000 hours

monthly) into text to enable customer sentiment analysis in those calls.

From this AI project, NoBroker wants to find out how customers feel about existing services and what they want to see next. This machine learning algorithm requires complex datasets consisting of deep speech architecture with around 38 million parameters, 260 hours of Indian speech, and over 6,700 lines of Python code.

“We faced performance limitations running AI workflows in our legacy system. A major obstacle was CPU speed,” says Akhil. “Using GPUs on [Compute Engine \(/compute\)](#), our data scientists can run machine learning training and evaluation in minutes instead of days.”

GPUs on Compute Engine, capable of up to 960 teraflops of performance per instance, are ideal for compute-intensive tasks such as machine learning and AI.

“We activate GPUs only when training is needed. Once the training job is complete, the administrator stops the instance on Google Cloud, so we only pay for what we use,” says Akhil.



Delivering rich location details with Google Maps Platform

NoBroker uses Google Maps Platform APIs and SDKs to deliver rich location insights when customers search for properties. Developers integrate NoBroker's MySQL database with Google Maps to display properties for sale or for rent on the app.

"Customers can access their app to check out the neighborhood ahead of a physical visit," says Akhil. "They can zoom in to see nearby points of interest and services such as schools and parks." The Places API

(<https://developers.google.com/places/web-service/intro>) provides rich detail for more than 150 million points of interest in the world.

The Directions API

(<https://developers.google.com/maps/documentation/directions/start>)

suggests the best route from Point A to Point B. Whether they're traveling by foot, by car or by public

transportation, the customer knows exactly where to go without getting lost. The [Distance Matrix API](https://developers.google.com/maps/documentation/distance-matrix/) (<https://developers.google.com/maps/documentation/distance-matrix/>) informs the customer of the commute time, not just the distance, since traveling short distances can take a long time during peak hours.

To save time on property searches, customers can type in the street address or drop a marker on Google Maps to indicate the desired property location. The [Geocoding API](https://developers.google.com/maps/documentation/geocoding/start) (<https://developers.google.com/maps/documentation/geocoding/start>) automatically converts latitude and longitude coordinates into street addresses on Google Maps.

Optimizing site performance with microservices and Cloud CDN

The majority of NoBroker's business occurs on its digital platform. NoBroker wants to improve the app to attract and retain customers. If too many users are on the site, the browser loads slowly to retrieve content from the web server.

"To speed up loading time for customers, we're rewriting our UI front end as a modern microservices-based application that serves content from [Cloud Storage \(/storage\)](#)," says Akhil. "We're using [Cloud CDN \(/cdn\)](#) to cache static page files such as Javascript, CSS, and images in edge servers closer to the user."

NoBroker uses AJAX as the communication protocol between the web application and REST API server on Google Cloud.

For example, when someone from New York visits NoBroker.com, a cache version of the static page is rendered on their browser, while the rest of the page loads in the background. This approach dramatically reduces the latency of user requests traveling thousands of miles to NoBroker's web server in India. Akhil notes that the page starts loading within 40 to 200 milliseconds.

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A data lake of valuable insights on Starship

NoBroker is building a data lake called Starship on Cloud Storage to ingest and store data to help it come up with more insightful tools, to make a customer’s real estate search more seamless and intuitive, for example. NoBroker will also be able to better understand what consumers want so that it can provide them with faster and more accurate recommendations, rent predictions, and location or property suggestions.

From an engineering perspective, the Starship project makes use of open source technology to better integrate with third-party apps and provide more options for gathering insights from data to improve NoBroker’s features, products, and more. NoBroker readily shares its real estate insights with customers in the form of reports such as [India Rental Habits and Trends](#)

(<https://www.nobroker.in/blog/indian-rental-habits-and-trends/>)

and [Trending Localities to Invest In - Mumbai 2020](https://www.nobroker.in/blog/trending-localities-to-invest-in-mumbai/)

(<https://www.nobroker.in/blog/trending-localities-to-invest-in-mumbai/>)

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“We want to create a single point of data on Google Cloud for better recommendations for our customers,” says Akhil. “Big data analysis helps us spot trends such as where people are buying and when owners are planning to sell. With the help of Google Cloud, we can push forward as an AI-first real estate platform and keep improving our machine learning models to serve our customers better.”

